

Force and extension (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Investigating how force affects the extension of a spring: name the independent variable.	Force applied to the spring	☐
2 ★	Investigating how force affects the extension of a spring: name the dependent variable.	Extension of the spring	☐
3	State the extension of a spring when no force is applied.	Zero	☐
4 ★	Describe how to convert the mass hung on a spring into a weight.	Multiply the mass in kg by the gravitational field strength (9.8 N/kg).	☐
5	A student records the whole length of a spring, not its extension. Name the type of error this causes.	A systematic error (every value is too large).	☐
6	The force on a spring is doubled. State the effect on its extension.	It doubles (below the limit of proportionality).	☐